Contents

- Chapter 1: Transfer function and stability properties
- Chapter 2: Feedback stabilization of the bicycle
- Chapter 3: Root locus, self-stability and self-aligning speed
- Chapter 4: Control of the Whipple bicycle model
- Final comparison

Chapter 1: Transfer function and stability properties

The simplified bicycle model is obtained by considering small roll and steering angles during straight motion. The main output is the roll angle φ, while the steering angle δ is used as the control input. The bicycle and rider are treated as a rigid body with mass m, centre of mass height h, and geometric parameters a and b. With this approximation, the dominant effects are gravity and the velocity-dependent steering terms.

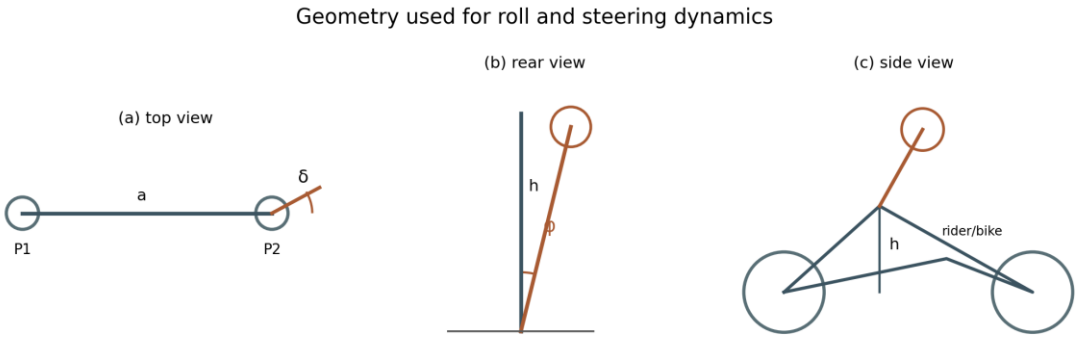


Figure 1. Parameters used for the simplified bicycle steering dynamics.

The roll equation can be expressed as

$$J\,\ddot{\varphi} - (D\,v0 / b)\,\dot{\delta} = m\,g\,h\,\sin(\varphi) + (m\,v0^2\,h / b)\,\delta$$

For small tilt angles, $\sin(\varphi) \approx \varphi$. Taking the Laplace transform with zero initial conditions gives the transfer function from steering angle to roll angle:

$$G\varphi\delta(s) = \Phi(s) / \Delta(s) = [(D\,v0 / b)\,s + (m\,v0^2\,h / b)] / [J\,s^2 - m\,g\,h]$$

The high-frequency gain and zero-frequency gain are therefore

$$k' = D\,v0 / (b\,J), \quad k0 = G\varphi\delta(0) = -v0^2 / (b\,g)$$

The zero and poles are obtained as

$$z1 = -m\,v0\,h / D, \quad p1,2 = \pm\,\mathrm{sqrt}(m\,g\,h / J)$$

For forward motion, $v0 > 0$, the zero is located in the left half-plane, so the plant is minimum phase. However, one pole is positive and one pole is negative. Consequently, the open-loop bicycle roll dynamics are unstable, even though the zero is stable. The relative degree is one because the denominator order is two and the numerator order is one.

Chapter 2: Feedback stabilization of the bicycle

A first stabilizing idea is to use a proportional steering feedback law based on the measured roll angle:

$$\delta = -kp\,\varphi$$

Substituting this relation in the small-angle roll equation gives the closed-loop characteristic equation

$$J\,\ddot{\varphi} + (D\,v0\,kp / b)\,\dot{\varphi} + [(m\,v0^2\,h\,kp / b) - m\,g\,h]\,\varphi = 0$$

For this second-order polynomial to have both roots in the open left half-plane, all coefficients must have the same sign. Therefore, the proportional gain must satisfy

$$kp > b\,g / v0^2$$

This condition shows the main limitation of pure steering feedback: as the forward velocity approaches zero, the required gain grows without bound. At zero speed the steering mechanism cannot generate the same stabilizing dynamic effect, since the stabilizing terms are proportional to v0 or $v0^2$.

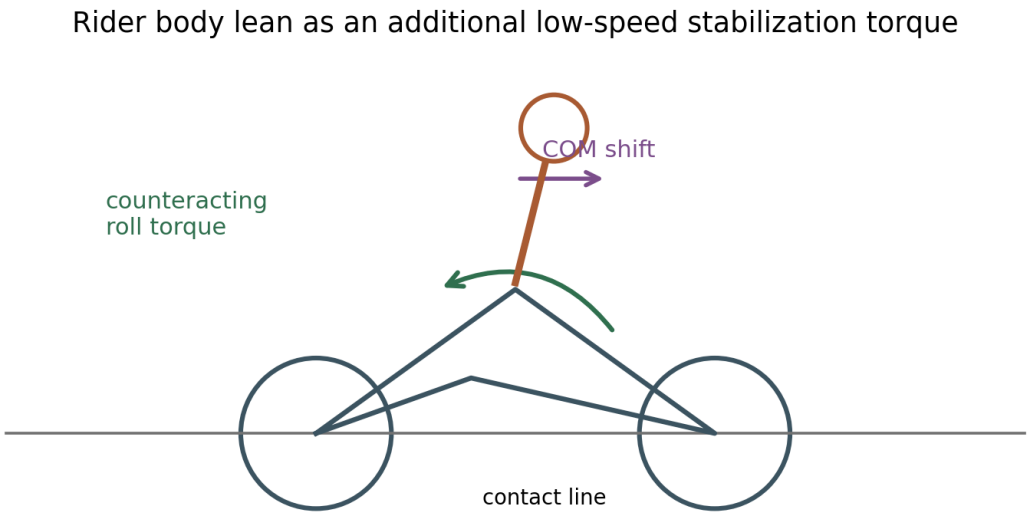


Figure 2. Rider body lean as a direct roll-torque action at very low speed.

To model the rider action at very low speed, an additional counter-reacting roll torque can be introduced:

$$\tau r = - kr\, \varphi - cr\, \dot{\varphi}$$

The resulting low-speed characteristic terms remain controllable even when v0 is nearly zero. In particular, at v0 = 0 the stabilizing conditions become

$$cr > 0, \quad kr > m\, g\, h$$

Thus, rider body lean provides a stabilizing mechanism that acts directly on roll dynamics and does not disappear when the bicycle stops.

Chapter 3: Root locus, self-stability and self-aligning speed

For the Whipple bicycle model, the linearized roll-steer dynamics around upright motion are written in matrix form as

$$M\, \ddot{q} + C\, v0\, \dot{q} + (K0 + K2\, v0^2)\, q = [0, T]^T, \quad q = [\varphi, \delta]^T$$

After multiplying by M^{-1} and defining the state vector $x = [\varphi, \delta, \dot{\varphi}, \dot{\delta}]^T$, the autonomous dynamics are

$$\dot{x} = A(v0)\, x$$

$$A(v0) = [[0_2 x_2, I_2 x_2], [-M^{-1}(K0 + K2\, v0^2), -M^{-1}\, C\, v0]]$$

The real parts of the four eigenvalues determine the stability range. A positive real part corresponds to an unstable mode, while negative real parts for all modes indicate self-stable motion.

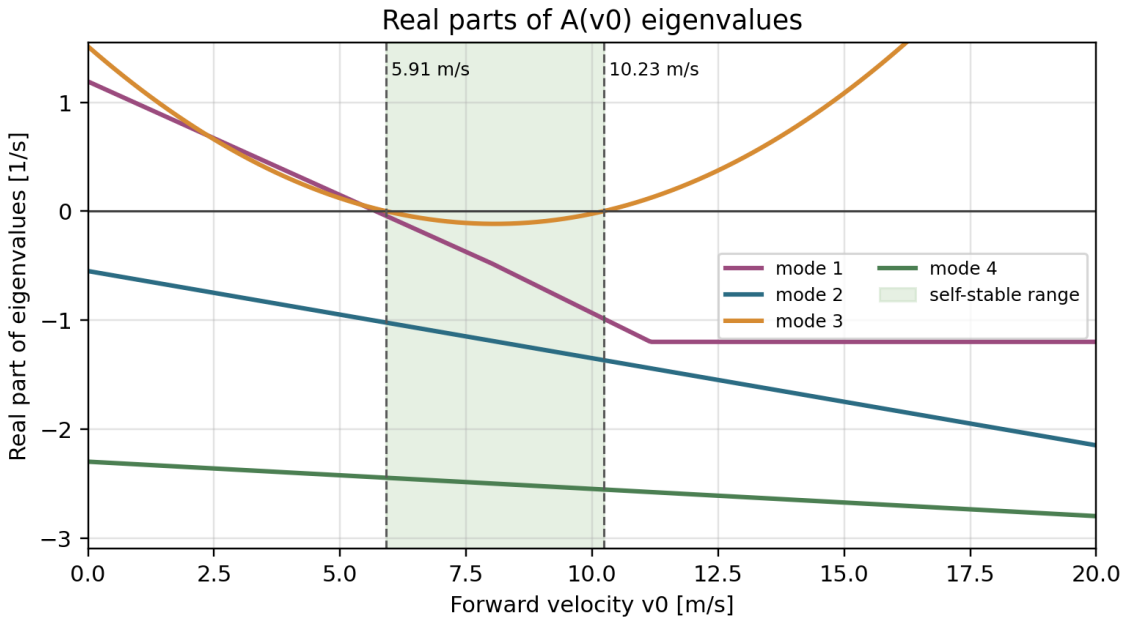


Figure 3. Real parts of the four eigenvalues as a function of forward speed.

The plot indicates that the bicycle is unstable at low speed, becomes self-stable in the intermediate interval from approximately 5.91 m/s to 10.23 m/s, and loses stability again at higher speed due to a capsize-type mode.

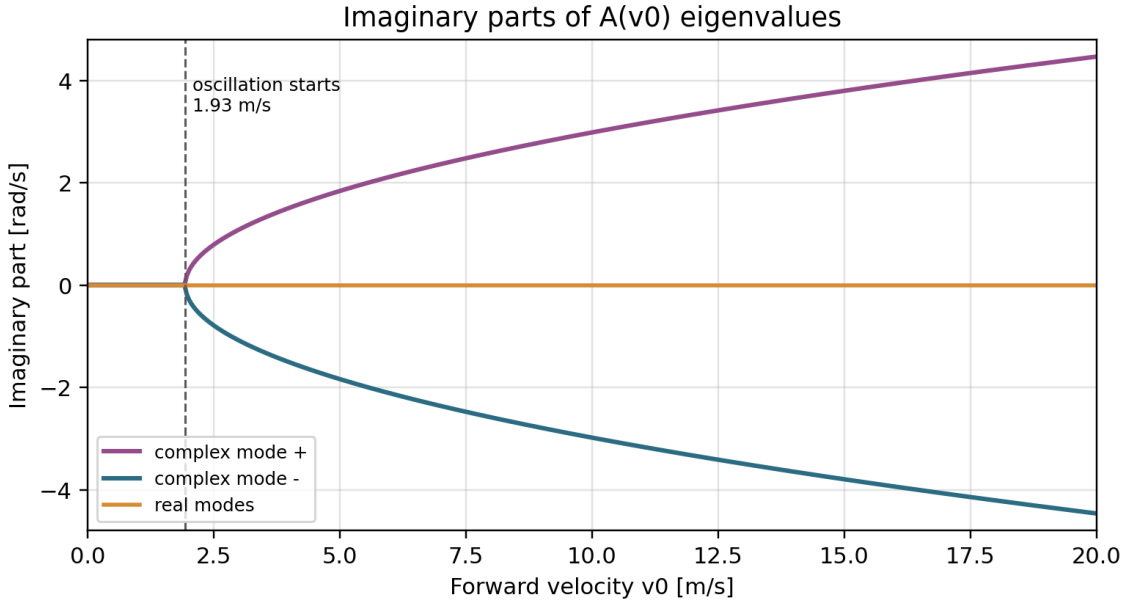


Figure 4. Imaginary parts of the eigenvalues as a function of forward speed.

Oscillatory behavior starts when the eigenvalues become complex. In this reconstruction, the first nonzero imaginary pair appears at approximately 1.93 m/s. The remaining real modes keep zero imaginary part.

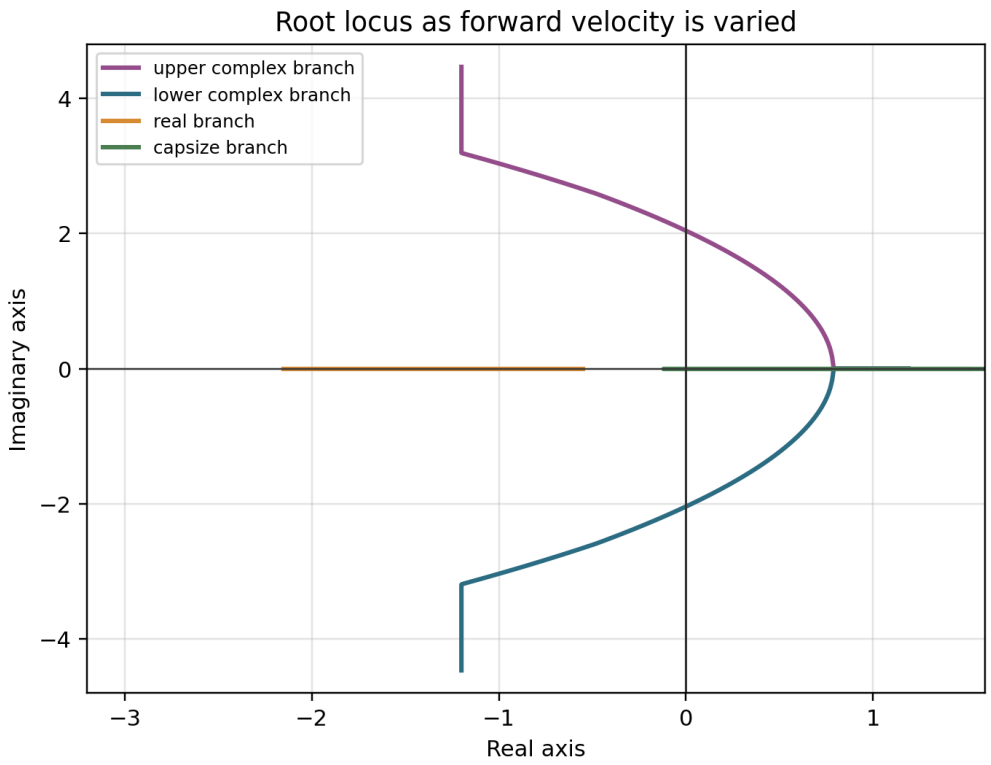


Figure 5. Root-locus representation of the velocity-dependent eigenvalues.

The self-aligning speed is the minimum forward velocity for which every eigenvalue is in the left half-plane. From the eigenvalue plots, this value is

$$v_{\text{self-aligning}} \approx 5.91 \text{ m/s}$$

Chapter 4: Control of the Whipple bicycle model

The final part considers a controlled Whipple bicycle model. The input is the steering torque T and the state vector is $x = [\varphi, \delta, \dot{\varphi}, \dot{\delta}]^T$. The state-space representation is written as

$$\dot{x} = A\,x + B\,u, \quad y = C\,x$$

$$C_\varphi = [1\ 0\ 0\ 0], \quad C_\delta = [0\ 1\ 0\ 0]$$

First, a full-state feedback controller is designed using Ackermann pole placement. The requested closed-loop poles are -2, -10 and -1 ± i. Using the reachability matrix R = [B, AB, A²B, A³B], the state-feedback gain is

$$K_c = [0.5125,\ 1.2736,\ -3.5599,\ 0.3493]$$

A feedforward term is included to track a constant steering reference $\delta_{ref} = 0.002$ rad:

$$u = K_c\,x + k_{ff}\,r, \quad k_{ff} = -1 \,/\, (C_\delta\,(A + B\,K_c)^{-1}\,B)$$

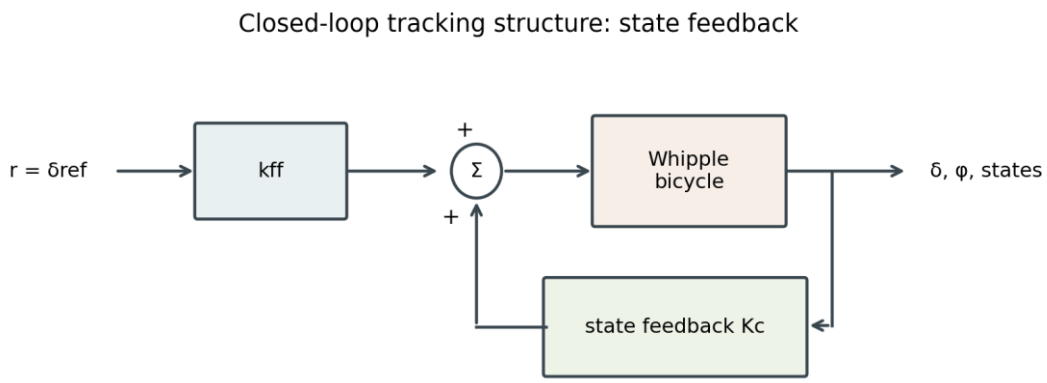


Figure 6. Closed-loop structure for state feedback with feedforward reference action.

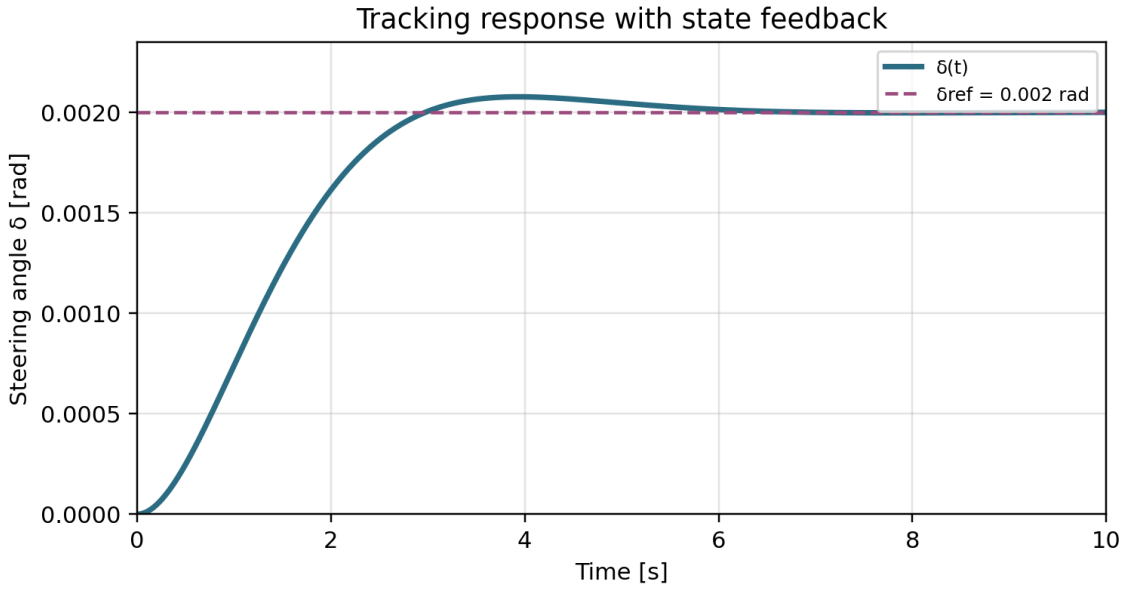


Figure 7. Steering-angle response using state feedback.

The response reaches the desired steering angle with a well-damped transient. This result is consistent with the assigned closed-loop pole locations and the relatively high damping imposed by the complex pair -1 ± i.

The second controller is a proportional output-feedback law based only on the roll-angle measurement:

$$u = kp\,\varphi + kff\,r$$

A search over possible proportional gains shows that it is not possible to satisfy the settling-time and damping-ratio requirements simultaneously. A compromise gain is chosen as

$$kp = 3.98, \quad \zeta min = 0.2684$$

$$\text{Closed-loop poles: } -11.8792,\ -1.5358 \pm 5.512\,i,\ -1.001$$

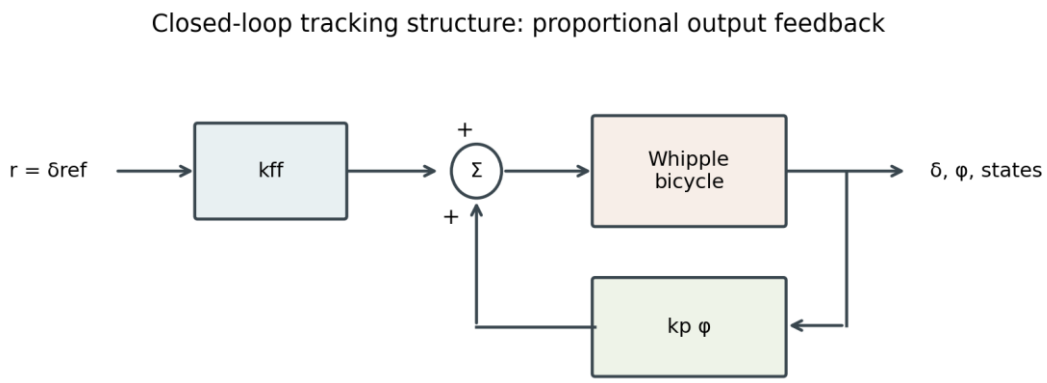


Figure 8. Closed-loop structure for proportional output feedback with feedforward action.

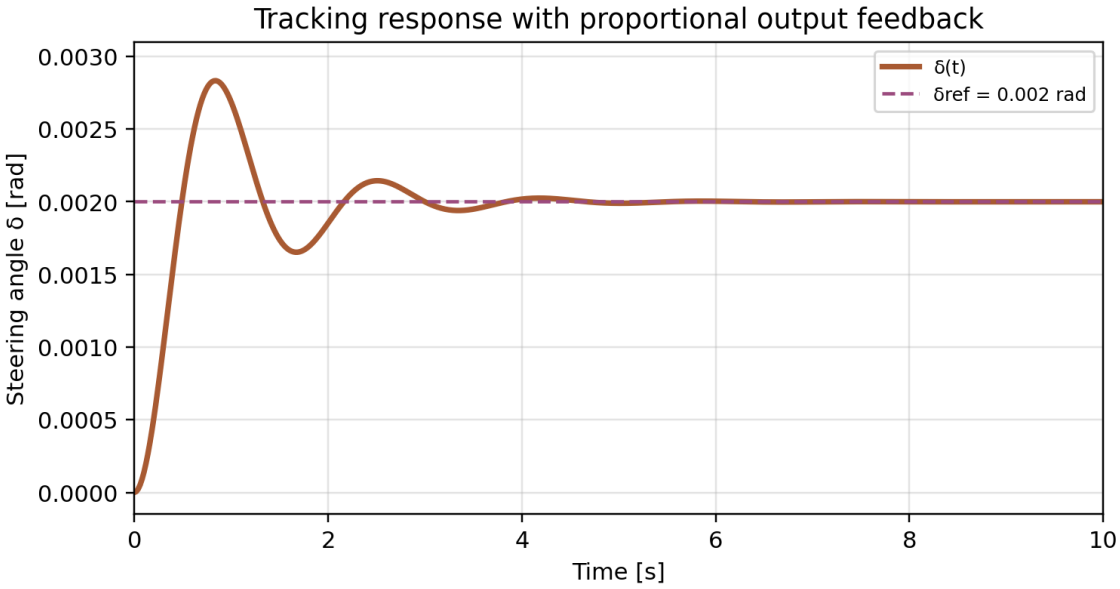


Figure 9. Steering-angle response using proportional output feedback.

Final comparison

Both controllers stabilize the closed-loop bicycle model and make the steering angle converge to the desired reference value. However, their dynamic performances are different. The full-state feedback design uses all state variables and can place the closed-loop poles exactly at the desired locations. This gives a faster and more damped response.

The proportional output-feedback controller is simpler because it uses only the tilt angle. Nevertheless, its single gain is not sufficient to independently tune all four closed-loop poles. The best gain found for the required settling condition still gives a low damping ratio, which leads to oscillation and larger overshoot. Therefore, state feedback provides the better control performance, while proportional output feedback remains easier to implement but less satisfactory from a comfort and robustness point of view.